HW-7

① (a) $y_1 = e^x$; $y_2 = xe^x$

$$W(x) = \begin{vmatrix} e^x & xe^x \\ e^x & e^x(1+x) \end{vmatrix}$$

$$= \{e^x\} \begin{vmatrix} 1 & x \\ 1 & 1+x \end{vmatrix} = \{e^x\}(1+x-x)$$

$$= e^x \neq 0.$$

$W(x) \neq 0 \longrightarrow y_1$ & y_2 are LI (Proved)

(b) Choose $y_3(x) = x^2 e^x$.

$$W(x) = \begin{vmatrix} e^x & xe^x & x^2e^x \\ e^x & e^x(1+x) & e^x(x^2+2x) \\ e^x & e^x(1+x+1) & e^x(x^2+2x+2) \end{vmatrix}$$

$$= e^x \begin{vmatrix} 1 & x & x^2 \\ 1 & 1+x & x^2+2x \\ 1 & 2+x & x^2+4x+2 \end{vmatrix}$$

$$= e^x \begin{vmatrix} 1 & x & x^2 \\ 0 & 1 & 2x \\ 0 & 1 & 2x+2 \end{vmatrix} = e^x(2x+2-2x)$$

$$= 2e^x \neq 0.$$

$W(x) \neq 0 \longrightarrow y_1, y_2$ & y_3 are LI.

(2) $x > 0$

$$w(x) = \begin{vmatrix} e^{-1/x^2} & 0 \\ e^{-1/x^2} \cdot \frac{2}{x^3} & 0 \end{vmatrix} = 0.$$

$w(x) = 0 \longrightarrow$ Maybe LI or LD.

$x < 0$

$$w(x) = \begin{vmatrix} 0 & e^{-1/x^2} \\ 0 & e^{-1/x^2} \frac{2}{x^3} \end{vmatrix} = 0. \longrightarrow \text{Maybe LI/LD.}$$

$x > 0$:

$c_1 f + c_2 g = 0.$

$\Rightarrow c_1 e^{-1/x^2} = 0 \Rightarrow \boxed{c_1 = 0}$; ~~c_2 is arbitrary.~~

$x < 0$:

$c_1 f + c_2 g = 0.$

$\Rightarrow c_2 e^{-1/x^2} = 0 \Rightarrow \boxed{c_2 = 0}$

$(c_1, c_2) = (0,0) \longrightarrow f, g$ are LI. on $\mathbb{R}$. (Ans)

(3) $w(x) = w(2) \exp\left[-\int_2^x P(x')\,dx'\right]$

$P(x) = \frac{2}{x}$; $\int P(x)\,dx = 2\ln x = \ln x^2$

$\therefore \exp\left[-\int^x P(x')\,dx'\right] = \frac{1}{x^2}$

$\therefore w(x) = \frac{1}{x^2} \neq 0$

Hence y_1 & y_2 can never become dependent for $x > 0$

(4) $y_2 = y_1 \int \frac{\exp\left[-\int^x P(x')\,dx'\right]}{y_1^2}\,dx$

$P(x) = \frac{2}{x}$; $\exp\left[-\int^x P(x')\,dx'\right] = \frac{1}{x^2}$

$\therefore y_2 = \frac{\sin kx}{x} \int \frac{1/x^2 \cdot x^2}{\sin^2 kx}\,dx = \frac{\sin kx}{x} \int \operatorname{cosec}^2 kx\,dx$

$$= \frac{\sin kx}{x} \cdot \left(-\frac{\cot kx}{k}\right)$$

$$= \frac{1}{k}\,\frac{\cancel{\sin kx}}{x} \cdot \frac{\cos kx}{\cancel{\sin kx}} = -\frac{\cos kx}{kx}$$

$$\therefore \boxed{y_2(x) = -\frac{\cos kx}{kx}} \quad \text{(Ans)}$$

(5) $p(x) = \frac{1}{x} \longrightarrow$ NON-ANALYTIC @ $x=0$. (Pole of order 1)

$Q(x) = 1$

$x=0 \longrightarrow$ Regular Singular Point.

$$y(x) = \sum_{n=0}^{\infty} a_n x^{n+r}$$

$$y'(x) = \sum_{n=0}^{\infty} (n+r)\, a_n x^{n+r-1}$$

$$y''(x) = \sum_{n=0}^{\infty} (n+r)(n+r-1)\, a_n x^{n+r-2}$$

Substituting into the ODE:

$$\sum_{n=2}^{\infty} (n+r)(n+r-1)\, a_n x^{n+r} + \sum_{n=1}^{\infty} (n+r)\, a_n x^{n+r} + \sum_{n=0}^{\infty} a_n x^{n+r+2} = 0$$

The lowest power of x is x^r (@ $n=0$): Coefficient of x^r must vanish for indicial eq^n:

$$\therefore \quad r(r-1)\, a_0 \cancel{x^r} + r a_0 = 0.$$

$$\Rightarrow \quad r^2 = 0 \quad \Rightarrow \quad \boxed{r = 0, 0}.$$

$$\therefore \sum_{n=2}^{\infty} n(n-1)\, a_n x^{n+r} + \sum_{n=2}^{\infty} n a_n x^{n+r} + \underbrace{\sum_{n=2}^{\infty} a_{n-2} x^{n+r}}_{[n+2 \to n]} = 0 \quad [r=0].$$

$$+ a_1 x +$$

Coefficient of x^n: $[n(n-1) + n]\, a_n + a_{n-2} = 0.$

$$\Rightarrow \quad a_n = \frac{-a_{n-2}}{n^2}$$

$$\Rightarrow \quad \boxed{a_{n+2} = -\frac{a_n}{(n+2)^2}} \quad [n \to n+2]$$

Coefficient of x: $\boxed{a_1 = 0}$

$n=0$: $a_2 = -\frac{a_0}{4}$

$n=1$: $a_3 = -a_1/9 = 0.$

$n=2$: $a_4 = \frac{-a_2}{16} = \frac{a_0}{64}.$

$n=4$: $a_6 = -\frac{a_4}{36} = -\frac{a_0}{36 \times 64}$

$$\therefore\ y(x) = \sum_{n=0}^{\infty} a_n x^n$$

$$= a_0\left[1 - \frac{x^2}{4} + \frac{x^4}{64} - \dots\dots\right]$$ (all odd coefficients $= 0$ as $a_1 = 0$)

choose $a_0 = 1$:

$$\boxed{J_0(x) = 1 - \frac{x^2}{4} + \frac{x^4}{64} - \dots}$$ (Ans)

$$y_2(x) = y_1 J_0 \int \frac{\exp\left[-\int^x P(x')\,dx'\right]}{J_0^2}\,dx.$$

$$= J_0 \int \frac{1/x}{J_0^2}\,dx.$$

$$= J_0 \int \frac{1}{x}\left(1 - \frac{x^2}{4} + \frac{x^4}{64} - \dots\right)^{-2} dx.$$

$$= J_0 \int \frac{1}{x}\left(1 + \frac{x^2}{2} + \frac{5x^4}{32} + \dots\right) dx.$$

$$= J_0 \int \left(\frac{1}{x} + \frac{x}{2} + \frac{5x^3}{32} + \dots\right) dx.$$

$$= J_0\left[\ln x + \frac{x^2}{4} + \frac{5x^4}{128} + \dots\right]$$

$$\boxed{y_2(x) = J_0(x)\ln x + \cancel{J_0(x)}\ \text{power series}}$$ (Ans)

(6) $\cancel{w(x)}$ $y'' + P(x)y' + Q(x)y = 0 \longrightarrow y_1$, $\longrightarrow y_2$.

$$w(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1' \neq 0 \quad [\text{LI}].$$

Let $y_1(a) = y_1(b) = 0$ & y_1 has no other zeroes in $[a, b]$

$\therefore\ y_1'(a) \neq 0$; $y_1'(b) \neq 0$. else $y_1(x) = 0$. [uniqueness theorem] (contradiction)

~~Why?~~ ~~If $y_1'(a) = 0$ $\Rightarrow y_1''(a) = 0$ [from ODE]~~.

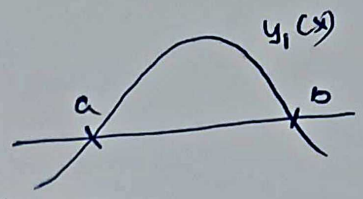
$\therefore$ & $w(a) = -y_2(a)\,y_1'(a)$.

$w(b) = -y_2(b)\,y_1'(b)$

$w(x)$ has the same sign over the entire interval.

$$\therefore\ \Big[\underbrace{w(x)}_{\text{retains sign of } w(a)} = \underbrace{w(a)}\ \underbrace{\exp\left[-\int^x P(x')\,dx'\right]}_{>0}$$

$\because$ y_1 crosses x axis @ a & b, signs of $y_1'(a)$ & $y_1'(b)$ are opposite.



~~$\therefore S[w(a)] = S[w(b)]$~~ ~~$S \rightarrow$ gives sign of a function~~

$\therefore$ $y_2(a)$ & $y_2(b)$ must have opposite signs if $w(a)$ & $w(b)$ must have the same sign.

$\therefore$ y_2 has a root ~~b/w~~ in $[a, b]$

$\therefore$ y_2 has a root in b/w 2 roots of y_1 (proved)

(7) $P(x) = -\dfrac{2x}{1-x^2}$; $Q(x) = \dfrac{2}{1-x^2}$. (18)

The singular pts. ~~of~~ are $x = \pm 1$.

$x = 1 \longrightarrow$ pole of order 1 for $P(x)$ & $Q(x)$.

$x = -1 \longrightarrow$ " " " " "

$x = \pm 1 \longrightarrow$ Regular Singular Points.